Status	Started	Completed
Finished	Wednesday, 11 November 2026, 1:05 PM	Wednesday, 11 November 2026, 1:07 PM

Duration	Grade
1 min 4 secs	**20.00** out of 20.00 (**100**%)

Question 1

Flag question

Correct Mark 1.00 out of 1.00

Your tool schema for create_ticket has a required priority field with enum values: ['low', 'medium', 'high', 'critical']. In production, tickets occasionally arrive needing 'urgent' treatment that falls between high and critical. The model silently assigns 'high' instead. What schema change handles new priority levels without silent misclassification?

a. Add 'urgent' to the enum — keep adding values as new levels emerge.
b. Create a separate create_urgent_ticket tool.
c. Add 'other' to the enum with a companion optional priority_detail string field. Also include a prompt instruction and few-shot example showing edge cases being classified as 'other' with a description rather than forced into a wrong existing category. ✓
d. Remove the enum constraint and use a free-text priority field.

The correct answer is: Add 'other' to the enum with a companion optional priority_detail string field. Also include a prompt instruction and few-shot example showing edge cases being classified as 'other' with a description rather than forced into a wrong existing category.

Question 2

Flag question

Correct Mark 1.00 out of 1.00

Your search tool returns results with a relevance_score field (0.0 to 1.0) but all scores cluster between 0.7 and 0.9 regardless of actual query relevance. The agent treats all results as equally relevant. What is the tool design issue?

a. Normalise all scores to a wider range in the tool response post-processing.
b. Add a threshold parameter so agents can filter by minimum relevance score.
c. The relevance scores are not calibrated to meaningfully differentiate results. The tool should return scores that reflect actual relevance variance — using proper scoring algorithms. If true calibration is not achievable, remove the field rather than providing misleading uniformly-high scores that give the agent false confidence in result quality. ✓
d. Remove the relevance_score field since it provides no signal.

The correct answer is: The relevance scores are not calibrated to meaningfully differentiate results. The tool should return scores that reflect actual relevance variance — using proper scoring algorithms. If true calibration is not achievable, remove the field rather than providing misleading uniformly-high scores that give the agent false confidence in result quality.

Question 3

Flag question

Correct Mark 1.00 out of 1.00

Your tool schema has a field document_language with no constraints. The agent fills this with full locale descriptions like 'this appears to be written in British English' instead of standard codes. Downstream systems expect ISO 639-1 codes (en, fr, de). What schema fix enforces the correct format?

a. Add a prompt instruction: 'Use ISO 639-1 language codes for document_language.'
b. Add a validation step after extraction that maps verbose descriptions to ISO codes.
c. Change document_language to an enum constraint listing all valid ISO 639-1 codes, or add a pattern constraint enforcing the two-letter lowercase format: {"type": "string", "pattern": "^[a-z]{2}$"}. Schema-level constraints are more reliable than prompt instructions for format enforcement. ✓
d. Accept free-text language descriptions — downstream systems can normalise them.

The correct answer is: Change document_language to an enum constraint listing all valid ISO 639-1 codes, or add a pattern constraint enforcing the two-letter lowercase format: {"type": "string", "pattern": "^[a-z]{2}$"}. Schema-level constraints are more reliable than prompt instructions for format enforcement.

Question 4

Flag question

Correct Mark 1.00 out of 1.00

Your agent calls a search tool that returns 200 results. Processing 200 results consumes most of the context window, leaving little room for the agent's reasoning. The tool has no result limiting capability. What is the most impactful tool schema improvement?

a. Increase the context window to accommodate 200 results.
b. Post-process the 200 results to return only the top 10 after the tool call.
c. Add a max_results parameter to the tool schema with a sensible default (e.g. 10) and maximum (e.g. 50), and add relevance_threshold as an optional filter. This prevents default over-retrieval, reduces context consumption by 80-90%, and leaves the model's context available for reasoning rather than data storage. ✓
d. Instruct the agent to only read the first 10 results from the tool's output.

The correct answer is: Add a max_results parameter to the tool schema with a sensible default (e.g. 10) and maximum (e.g. 50), and add relevance_threshold as an optional filter. This prevents default over-retrieval, reduces context consumption by 80-90%, and leaves the model's context available for reasoning rather than data storage.

Question 5

Flag question

Correct Mark 1.00 out of 1.00

Your agent uses a data transformation tool that internally calls three different services. When the third service fails, the tool returns an error but the changes from services 1 and 2 have already been applied. The agent retries the entire tool call, creating duplicate records from services 1 and 2. What tool design principle addresses this?

a. Catch the service 3 failure and proceed with partial success.
b. Add a rollback capability so the agent can undo partial completions before retrying.
c. Implement idempotency in the tool: either use a transaction pattern that rolls back all changes if any service fails, or include an idempotency_key parameter that services 1 and 2 use to detect and skip duplicate operations on retry. The agent should be able to safely retry without side effects.
d. The agent should never retry tools that call multiple services.

The correct answer is: Implement idempotency in the tool: either use a transaction pattern that rolls back all changes if any service fails, or include an idempotency_key parameter that services 1 and 2 use to detect and skip duplicate operations on retry. The agent should be able to safely retry without side effects.

Question 6

Flag question

Correct Mark 1.00 out of 1.00

You need to implement a two-phase workflow where phase 1 always extracts metadata and phase 2 enriches based on that metadata. Phase 2 must never run without successful phase 1 output. With tool_choice: auto, phase 2 occasionally runs first. What is the deterministic fix?

a. Use a PostToolUse hook to trigger phase 2 automatically after phase 1 completes.
b. Add extract_metadata as a required pre-condition in the enrich tool's schema.
c. On the first API call, set tool_choice: {type: 'tool', name: 'extract_metadata'} to force phase 1. After receiving and validating the metadata in application code, proceed to phase 2 on the subsequent API call. Programmatic sequencing in application code is deterministic; model-driven sequencing via prompts is probabilistic.
d. Use a system prompt: 'Always run phase 1 before phase 2 without exception.'

The correct answer is: On the first API call, set tool_choice: {type: 'tool', name: 'extract_metadata'} to force phase 1. After receiving and validating the metadata in application code, proceed to phase 2 on the subsequent API call. Programmatic sequencing in application code is deterministic; model-driven sequencing via prompts is probabilistic.

Question 7

Flag question

Correct Mark 1.00 out of 1.00

Your agent incorrectly uses Bash('python script.py') to run a Python analysis script instead of using the appropriate built-in tools for file analysis. A colleague defends this as 'getting the job done.' What is the correct position?

a. Bash and built-in tools are interchangeable — use whichever is more convenient.
b. Bash should only be used when no purpose-built built-in tool can accomplish the task. Using Bash for operations that built-in tools handle introduces shell-injection risks, produces less structured output, and bypasses the semantic clarity that tool descriptions provide for reliable agent decision-making. ✓
c. Bash is always preferable to built-in tools for technical tasks involving code execution.
d. Using Bash to run scripts is acceptable when no purpose-built tool exists for the task.

The correct answer is: Bash should only be used when no purpose-built built-in tool can accomplish the task. Using Bash for operations that built-in tools handle introduces shell-injection risks, produces less structured output, and bypasses the semantic clarity that tool descriptions provide for reliable agent decision-making.

Question 8

Flag question

Correct Mark 1.00 out of 1.00

Your tool processes a user-supplied file path and reads the file at that location. A security researcher discovers that passing '../../../etc/passwd' as the file path allows reading sensitive system files. What tool design principle prevents this?

a. Add a system prompt instruction: 'Never read files outside the project directory.'
b. Log all file path requests and monitor for suspicious patterns.
c. Implement path validation inside the tool: check that the resolved absolute path is within the permitted directory scope before reading. Reject any path that resolves outside the allowed boundaries. Input validation and path sanitisation are tool-layer responsibilities, not model-layer responsibilities. ✓
d. Use a read-only filesystem permission so sensitive files cannot be accessed.

The correct answer is: Implement path validation inside the tool: check that the resolved absolute path is within the permitted directory scope before reading. Reject any path that resolves outside the allowed boundaries. Input validation and path sanitisation are tool-layer responsibilities, not model-layer responsibilities.

Question 9

Flag question

Correct Mark 1.00 out of 1.00

Your tool receives requests to look up orders, but 20% of requests include customer names that match multiple accounts. The tool currently returns the first matching account arbitrarily. What is the correct tool response design?

a. Return all matching accounts with their full details and let the agent choose.
b. Return the most recently active account as the default match.
c. Return an error when multiple matches are found.
d. Return a structured multiple_matches response: {found: true, match_count: N, matches: {account_id, name, email, account_type}, requires_disambiguation: true}. The agent can then ask the user for the additional identifier needed to disambiguate rather than making an arbitrary selection that could cause errors. ✓

The correct answer is: Return a structured multiple_matches response: {found: true, match_count: N, matches: {account_id, name, email, account_type}, requires_disambiguation: true}. The agent can then ask the user for the additional identifier needed to disambiguate rather than making an arbitrary selection that could cause errors.

Question 10

Flag question

Correct Mark 1.00 out of 1.00

Your agent needs to find all TypeScript interface definitions across a 300-file repository to understand the data model. What is the most efficient sequence of built-in tools?

a. Read all 300 files sequentially and extract interface definitions manually.
b. Use Write to generate a search script that reads and analyses all TypeScript files.
c. Use Bash with find and grep in a single pipeline command.
d. Use Glob to find all .ts and .tsx files, then use Grep to search those files for the 'interface ' keyword pattern. Glob for file discovery by extension, Grep for content search within discovered files — each tool used for its designed purpose. ✓

The correct answer is: Use Glob to find all .ts and .tsx files, then use Grep to search those files for the 'interface ' keyword pattern. Glob for file discovery by extension, Grep for content search within discovered files — each tool used for its designed purpose.

Question 11

Flag question

Correct Mark 1.00 out of 1.00

Your tool schema marks transaction_id as a required field. The agent sometimes calls the tool without providing a transaction_id for cases where no transaction has occurred yet. The tool throws an error. How should the schema be designed to handle this?

a. Make transaction_id optional (nullable) in the schema — if no transaction has occurred, the tool should handle a null or absent transaction_id gracefully rather than requiring the agent to fabricate a transaction identifier. ✓

b. Add an enum with 'none' as a valid transaction_id value.

c. Change transaction_id to required with a default value of 'PENDING'.

d. Create a separate tool for cases without transaction IDs.

The correct answer is: Make transaction_id optional (nullable) in the schema — if no transaction has occurred, the tool should handle a null or absent transaction_id gracefully rather than requiring the agent to fabricate a transaction identifier.

Question 12

Flag question

Correct Mark 1.00 out of 1.00

Your codebase exploration agent finds a configuration file and needs to understand its structure. The file is a 200-line YAML config. Which built-in tool sequence should the agent use?

a. Use Glob to locate the file first, then Write to copy it to a temp location before reading.

b. Use the Read tool to load the full file content — Read is purpose-built for file reading and returns structured content the agent can reason about without shell command construction or injection risk. ✓

c. Bash('cat config.yaml') to read the file content.

d. Use Grep to search for specific keys within the configuration file.

The correct answer is: Use the Read tool to load the full file content — Read is purpose-built for file reading and returns structured content the agent can reason about without shell command construction or injection risk.

Question 13

Flag question

Correct Mark 1.00 out of 1.00

Your MCP tool for code review returns findings as a plain text paragraph: 'Found 3 issues: the function on line 45 lacks error handling, the variable name is unclear, and the return type is incorrect.' The CI system cannot parse this reliably. What tool response redesign is needed?

a. Use a regex post-processor on the CI side to extract issue data from prose.
b. Redesign the tool response as a structured JSON array with typed fields: [{line: 45, severity: 'high', category: 'error_handling', message: '...'}, ...]. Structured responses with consistent field types enable reliable downstream parsing without text processing heuristics. ✓
c. Add 'please format as JSON' to the tool's input prompt.
d. Instruct the agent to reformat the tool's text response before returning it to the CI system.

The correct answer is: Redesign the tool response as a structured JSON array with typed fields: [{line: 45, severity: 'high', category: 'error_handling', message: '...'}, ...]. Structured responses with consistent field types enable reliable downstream parsing without text processing heuristics.

Question 14

Flag question

Correct Mark 1.00 out of 1.00

Your tool schema marks all 12 fields as required. In practice, 4 of the 12 fields are only relevant for specific document types (e.g. invoice_number is only relevant for invoices). When processing contracts, the model fabricates invoice numbers to satisfy the required constraint. What schema design principle resolves this?

a. Only mark fields as required when they are always present in all document types being processed. For document-type-specific fields, mark them as optional/nullable — type: ['string', 'null'] — allowing the model to return null when the field is genuinely absent rather than fabricating values. ✓
b. Accept fabricated values — they will not affect contract processing.
c. Add an enum with 'N/A' as a valid value for all optional fields.
d. Create separate schemas for each document type with only their relevant required fields.

The correct answer is: Only mark fields as required when they are always present in all document types being processed. For document-type-specific fields, mark them as optional/nullable — type: ['string', 'null'] — allowing the model to return null when the field is genuinely absent rather than fabricating values.

Question 15

Flag question

Correct Mark 1.00 out of 1.00

Your agent uses Write to add a docstring to the top of an existing 500-line module. After the operation, the developer reports that all 500 lines of existing code have been deleted. What caused this?

a. The Write tool has a bug that randomly truncates files.
b. Write replaces the entire file with whatever content is provided. The agent likely only provided the new docstring as the file content, overwriting all 500 lines. The Edit tool should be used for targeted changes — it modifies only the specified section while preserving all other file content. ✓
c. The file was open in another editor, causing a conflict.
d. Write requires the full file content to be provided but the agent did not read the file first.

The correct answer is: Write replaces the entire file with whatever content is provided. The agent likely only provided the new docstring as the file content, overwriting all 500 lines. The Edit tool should be used for targeted changes — it modifies only the specified section while preserving all other file content.

Question 16

Flag question

Correct Mark 1.00 out of 1.00

Your tool processes natural language queries and converts them into database queries. A user enters: 'find all customers'; ignore previous instructions. The tool executes the injection text as a literal database query extension. What defensive design is needed?

a. Add input validation that rejects requests containing the word 'ignore.'
b. Route all queries through a content moderation filter first.
c. Implement strict input sanitisation at the tool layer: validate and sanitise the natural language query before converting it to a database query. Never pass user input directly to query construction without validation. Parameterised queries and input normalisation prevent injection attacks regardless of the content. ✓
d. Add a system prompt warning about injection attempts.

The correct answer is: Implement strict input sanitisation at the tool layer: validate and sanitise the natural language query before converting it to a database query. Never pass user input directly to query construction without validation. Parameterised queries and input normalisation prevent injection attacks regardless of the content.

Question 17

Flag question

Correct Mark 1.00 out of 1.00

Your coding agent writes a new function to a source file using Write. The developer reports that comments, imports, and utility functions defined above the target location are now missing. What should the agent have used instead?

a. Create a new file for the function and import it into the original file.
b. Read the file first to understand its structure, then use Write to reconstruct it completely.
c. Bash with a heredoc to append the new function to the end of the file.
d. The Edit tool — it performs targeted modifications to specific sections while preserving all content outside the targeted area. Write replaces the entire file; for adding a function to an existing file, Edit is the correct tool. ✓

The correct answer is: The Edit tool — it performs targeted modifications to specific sections while preserving all content outside the targeted area. Write replaces the entire file; for adding a function to an existing file, Edit is the correct tool.

Question 18

Flag question

Correct Mark 1.00 out of 1.00

Your agent calls a document analysis tool that processes large PDFs. For a 200-page PDF, the tool takes 90 seconds and the agent blocks for the entire duration. A 10-page PDF takes 5 seconds. How should the tool be designed to handle this variance?

a. Add a page_count parameter so the agent knows how long to wait.
b. Split the tool into fast_analyse (≤20 pages) and full_analyse (>20 pages).
c. Use async polling: for requests expected to take more than a threshold (e.g. 10 seconds), return immediately with a job_id and provide a check_analysis_status tool. For fast requests, return synchronously. The agent can poll for long-running jobs while performing other tasks. ✓
d. Set a 90-second timeout universally to accommodate the worst case.

The correct answer is: Use async polling: for requests expected to take more than a threshold (e.g. 10 seconds), return immediately with a job_id and provide a check_analysis_status tool. For fast requests, return synchronously. The agent can poll for long-running jobs while performing other tasks.

Question 19

Flag question

Correct Mark 1.00 out of 1.00

A developer proposes adding a debug_mode parameter to production MCP tools so agents can get verbose diagnostic output during troubleshooting. What is the concern with this approach?

a. Debug parameters increase tool schema complexity unnecessarily.
b. Debug mode is acceptable as long as it is disabled by default.
c. Debug mode adds latency — it should only be used in development environments.
d. Debug mode in production tools can expose sensitive internal data, connection strings, stack traces, and system configuration in tool responses that become part of the agent's context. This is a security risk. Diagnostic information should be logged server-side, not returned to the agent. ✓

The correct answer is: Debug mode in production tools can expose sensitive internal data, connection strings, stack traces, and system configuration in tool responses that become part of the agent's context. This is a security risk. Diagnostic information should be logged server-side, not returned to the agent.

Question 20

Flag question

Correct Mark 1.00 out of 1.00

Your development team adds a new MCP tool monthly. Over 18 months, the agent's tool set has grown to 23 tools. You notice tool selection accuracy has declined from 95% to 71% over this period. What is the relationship between tool count and this decline?

a. Tool selection accuracy naturally improves as the model has more options to choose from.
b. Add longer descriptions to all 23 tools to improve selection accuracy.
c. Tool selection reliability degrades as tool count increases because the model must distinguish between more similar-looking descriptions. Above 4-5 tools per agent, decision complexity rises non-linearly. Distribute the 23 tools across specialised subagents, each with a focused tool set of 4-5 tools. ✓
d. The accuracy decline is unrelated to tool count — the model should be retrained.

The correct answer is: Tool selection reliability degrades as tool count increases because the model must distinguish between more similar-looking descriptions. Above 4-5 tools per agent, decision complexity rises non-linearly. Distribute the 23 tools across specialised subagents, each with a focused tool set of 4-5 tools.

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